

Mid-sem exam

● Graded

Student

Samyak Sanghvi

Total Points

19 / 20 pts

Question 1

Q1a

1 / 1 pt

✓ + 0.5 pts Contrast increases

✓ + 0.5 pts dark pixels become black, light pixels become white

+ 0 pts No/Wrong answer

Question 2

Q1b

1 / 1 pt

✓ + 1 pt Increase saturation of HSI

+ 1 pt (OR) Additive transformation in RGB

+ 0 pts No/Wrong answer

Question 3

Q1c

1 / 1 pt

✓ + 0.5 pts Correct: No

✓ + 0.5 pts Reason

+ 0 pts No/Wrong answer

Question 4

Q1d

1 / 1 pt

✓ + 0.5 pts Formula for Radon transform (or) mentioned: integral of f over line at distance p

✓ + 0.5 pts Final correct formulation: $2A\sqrt{r^2 - p^2}$ for $p < r$ and 0 otherwise

- 0.25 pts No/Wrong limits in the final formulation

+ 0 pts No/Wrong answer

Question 5

Q2a

2 / 2 pts

✓ + 0.5 pts i) True: Entries sum to 1

✓ + 0.5 pts i) Reason: Constant intensity should be unaffected

✓ + 0.5 pts ii) False: Some entries have to be negative

✓ + 0.5 pts ii) Reason: Smoothing if entries are non-negative

+ 0 pts No/Wrong answer

Question 6

Q2b

2 / 2 pts

✓ + 0.5 pts Unsharp mask formula: $g = f + k(f - h * f)$

✓ + 0.5 pts Mean filter formula: $h = 5 \times 5$, all $1/25$

+ 1 pt Correct Formula: $g = w * f$ where $w = \text{delta} + k(\text{delta} - h)$

✓ + 1 pt (Or) Correct Result: $w = \text{all } -k/25$ and middle entry $= (1+24k/25)$

+ 0 pts No/Wrong answer

Question 7

Q3a

2 / 2 pts

✓ + 0.5 pts Correct formula for $F(\mu, \nu)$

✓ + 1.5 pts Showing that $F(\mu, \nu) = F_1(\mu).F_2(\nu)$

+ 0 pts No/Wrong answer

Question 8

Q3b

2 / 2 pts

✓ + 0.5 pts Spatial Domain: $g_1 * (g_2 * f)$
Frequency Domain: $G_1.G_2.F$

✓ + 0.5 pts Spatial Gaussian: $k_i \cdot \exp(-t^2/(2 \cdot \sigma_i^2))$
Frequency Domain: $K_i \cdot \exp(-2 \cdot \pi^2 \cdot \sigma_i^2 \cdot \mu^2)$

✓ + 0.5 pts $G_3 = G_1.G_2 = K_1.K_2 \cdot \exp(-2 \cdot \pi^2 \cdot (\sigma_1^2 + \sigma_2^2) \cdot \mu^2)$

✓ + 0.5 pts Inverse FT of G_3 gives $\sigma = \sqrt{\sigma_1^2 + \sigma_2^2}$

+ 0 pts No/Wrong answer

Question 9

Q4a

2 / 2.5 pts

+ 1 pt $u = M/10 = 800/10 = 80, v = 0$

✓ + 0.5 pts (or) $u = 1/10$ using continuous FT

✓ + 0.5 pts Spikes at (80, 0) and (-80, 0), symmetry

✓ + 0.5 pts Notch filter formula: $H(u,v) = HP(u-80,v).HP(u+80,v)$

✓ + 0.5 pts Gaussian or Butterworth High Pass filter formula

+ 0 pts No/Wrong answer

Question 10

Q4b

1.5 / 1.5 pts

✓ + 0.5 pts Multiplication with H in frequency domain
Multiplication with W = convolution with w = mean filtering

✓ + 1 pt WHF = HWF i.e. order does not matter

+ 0 pts No/Wrong answer

Question 11

Q5

3.5 / 4 pts

✓ + 1 pt Calibration of noise using constant-intensity scene

✓ + 1.5 pts Calibration of PSF using known scene e.g. point source

- 0.5 pts Chosen scene's frequency spectrum is zero almost everywhere

- 0.5 pts Did not properly handle noise in measured PSF

✓ + 1 pt Formula for Wiener filter or other deconvolution

- 0.5 pts Inverse filter instead of practical deconvolution scheme

+ 0.5 pts Parameter setting for deconvolution

+ 1 pt Only performed denoising, no deblurring

+ 0 pts Incorrect or no answer

Name: Samyak Sanghvi

Entry no: 2023CS10807

COL7(6)83 Mid-Semester Exam

14 September, 2025

-
- Write your name and entry number on every sheet.
 - There are 5 questions. Each question is worth 4 marks.
 - Justify all answers. Answer without justification will not receive full marks, and may receive zero.
 - You are permitted to use one single-sided A4 sheet of handwritten notes.
-

Please sign the following declaration:

As a student of IIT Delhi, I will not give or receive aid in examinations. I will do my share and take an active part in seeing to it that others as well as myself uphold the spirit and letter of the Honour Code.

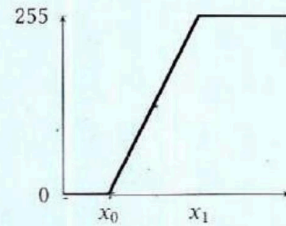
Signature: Samyak

1. Give a brief answer (with justification) for each of the following questions:

- (a) Consider the intensity transformation shown below, with $0 < x_0 < x_1 < 255$. What happens if it is repeatedly applied to an image?

$$T(x) = \begin{cases} 0 & x \leq x_0 \\ \frac{255}{x_1 - x_0} (x - x_0) & x_0 < x < x_1 \\ 1 & x \geq x_1 \end{cases}$$

after enough applications gap $\rightarrow 0$ it gives almost thresholding like effect.



$$T(x) \leq x_0 \text{ for } x \leq \left(\frac{x_1 - x_0}{255} + 1 \right) x_0$$

$$T(x) \geq x_1 \text{ for } x \geq x_1 \left(\frac{x_1 - x_0}{255} \right) + x_0$$

$$\Rightarrow \text{after enough application Gap} = \frac{(x_1 - x_0)^n}{255^n} \rightarrow 0$$

$$\therefore T^2(x) = \begin{cases} 0 & x \leq \left(\frac{x_1 - x_0}{255} + 1 \right) x_0 \\ \frac{255}{x_1 - x_0} \left(\frac{255}{x_1 - x_0} (x - x_0) - x_0 \right) & \text{else} \\ 1 & x \geq x_1 \left(\frac{x_1 - x_0}{255} \right) + x_0 \end{cases}$$

$$\frac{255}{x_1 - x_0} \left(\frac{255}{x_1 - x_0} (x - x_0) - x_0 \right) \text{ else}$$

$$1 \quad x \geq x_1 \left(\frac{x_1 - x_0}{255} \right) + x_0$$

$$\text{Gap} = \left(\frac{x_1 - x_0}{255} \right)^n (x_1 - x_0) \Rightarrow \text{thresholding like effect}$$

- (b) Explain how to enhance an RGB image which has good intensity contrast, but whose colours appear dull and faded and should be made more vibrant.

we must convert RGB to HSI domain

Since contrast is good we don't mess with I, we tune and increase Saturation and hue of via

histogram matching and try to make it more vibrant.

(c) Is bilateral filtering effective at removing salt-and-pepper noise? Why or why not?

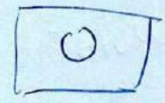
NO

Bilateral filtering works best for mostly similar neighbourhoods and identifies textures well.

Say on a pixel with salt ($I = I_{max}$) which was supposed to be some lower I value, bilateral filter will not decrease I , hence salt and pepper behaviour will mostly stay.
(as $e^{-\frac{(I_{max} - I_0)^2}{2\sigma^2}} < 1$)

(d) What is the Radon transform $g_0(\rho)$ of a disk function, $f(x, y) = \begin{cases} A & \text{if } x^2 + y^2 \leq r^2, \\ 0 & \text{otherwise?} \end{cases}$

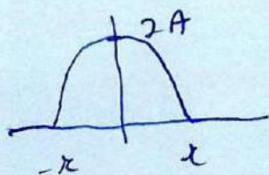
$$g_0(\rho) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x^2 + y^2) f(x, y) \delta(\rho - (x \cos \theta + y \sin \theta)) dx dy$$



due to symmetry $g_0(\rho) = g_{0^0}(\rho)$

$$\begin{aligned} \therefore g_0(\rho) &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) \delta(\rho - r) dx dy \\ &= \int_{-\infty}^{\infty} f(r, y) dy = \int_{-\sqrt{r^2 - \rho^2}}^{\sqrt{r^2 - \rho^2}} A dy \quad (\rho < r) \\ &\quad \text{else } 0 \end{aligned}$$

$$\therefore g_0(\rho) = \begin{cases} 2A \sqrt{r^2 - \rho^2} & (|\rho| \leq r) \\ 0 & \text{else} \end{cases}$$



2. Suppose we wish to design a spatial kernel w such that convolution with it sharpens an image; specifically, fine details are enhanced while the intensity in constant regions is unaffected.

(a) It is claimed that for such a kernel, (i) its entries should sum to 1, and (ii) all its entries should be nonnegative. Are these claims true? Justify your answer for both claims separately.

a) ~~false~~ True

consider a constant intensity i neighbourhood.

$$w * f(x, y) = \sum_s \sum_t w(s, t) f(x-s, y-t)$$

$$= i \sum_s \sum_t w(s, t) = i \text{ (constraint of question)}$$

$$\Rightarrow \sum_s \sum_t w(s, t) = 1$$

b) described kernel is a kernel for unsharp masking

$$g(x, y) = f(x, y) + K f_{HP}(x, y)$$

$$\therefore w_{uns} = I + K w_{HP}$$

$$\text{Laplace kernel (3x3)} = \begin{bmatrix} 1 & -8 & 1 \\ 1 & -8 & 1 \end{bmatrix} \text{ is a valid Highpass kernel}$$

$\therefore$ for $K = 4$, all entries in w_{uns} are not non-negative.

false

Alternative counter example: Q2b)

(b) Give the kernel that corresponds to unsharp masking with a 5×5 mean filter.

$$f_{LP} = w_{mean} * f$$

$$g(x,y) = f + K(f - f_{LP})$$

$$\Rightarrow f * I + K(I - w_{mean}) * f$$

$$\therefore w_{uns} = I + K(I - w_{mean})$$

$$I = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

$$w_{mean} = \frac{1}{25} \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \end{pmatrix}$$

$$\therefore w_{uns} = \frac{1}{25} \begin{bmatrix} -K & -K & -K & -K & -K \\ -K & -K & -K & -K & -K \\ -K & -K & 25+24K & -K & -K \\ -K & -K & -K & -K & -K \\ -K & -K & -K & -K & -K \end{bmatrix}$$

$$K \geq 1 \text{ (preferably).}$$

3. (a) Suppose a function f is separable, i.e. $f(t, z) = f_1(t)f_2(z)$, and the corresponding 1D continuous Fourier transforms are $f_1 \Leftrightarrow F_1, f_2 \Leftrightarrow F_2$. Using the definition of the 2D Fourier transform, find F such that $f(t, z) \Leftrightarrow F(\mu, \nu)$.

$$\begin{aligned}
 F(u, v) &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(t, z) e^{-i2\pi(u t + v z)} dt dz \\
 &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_1(t) e^{-2\pi i u t} \cdot f_2(z) e^{-2\pi i v z} dt dz \\
 &= \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} f_1(t) e^{-2\pi i u t} dt \right) f_2(z) e^{-2\pi i v z} dz \\
 &= \int_{-\infty}^{\infty} F_1(u) f_2(z) e^{-2\pi i v z} dz \\
 &= F_1(u) \int_{-\infty}^{\infty} f_2(z) e^{-2\pi i v z} dz \\
 &= F_1(u) \cdot F_2(v) \\
 \therefore f(t, z) &\Leftrightarrow F_1(u) \cdot F_2(v)
 \end{aligned}$$

- (b) Let g_1 and g_2 be Gaussian functions of standard deviation σ_1 and σ_2 respectively. A continuous signal f is smoothed by first convolving with g_1 and then with g_2 . Using the Fourier transform, show that the result is equivalent to convolving f once with a Gaussian, and find its standard deviation.

You may use the fact that $k \exp(-at^2) \Leftrightarrow K \exp(-\pi^2 \mu^2/a)$; the exact value of K shouldn't matter.

$$g = g_1 * g_2 * f$$

$$g = g_1 * g_2 * f$$

$$\therefore \cancel{G} = \cancel{G_1} \cdot \cancel{G_2} \cdot \cancel{F}$$

$$G(u) = G_1(u) \cdot G_2(u) \cdot F(u)$$

$$g_1 = K \exp(-x^2/2\sigma_1^2)$$

$$g_2 = K \exp(-x^2/2\sigma_2^2)$$

$$\therefore G_1 = K \exp(-2\sigma_1^2 \pi^2 u^2)$$

$$G_2 = K \exp(-2\sigma_2^2 \pi^2 u^2)$$

$$= K^2 \exp(-2(\sigma_1^2 + \sigma_2^2) \pi^2 u^2) \cdot \cancel{F} = \cancel{F(u)} \cdot F(u)$$

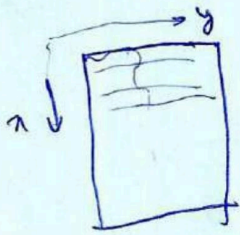
$$g(t) \Leftrightarrow K \cdot (K \exp(-x^2/2(\sigma_1^2 + \sigma_2^2)))^* f(t) = g(t)$$

$$\therefore g(t) = K g_3(t) * f(t)$$

Hence final effect is same as gaussian convolve with std deviation $\sqrt{\sigma_1^2 + \sigma_2^2}$

4. Suppose you are given a 800×600 image which is corrupted by sinusoidal horizontal stripes with a period of 10 pixels.

(a) At what frequency(ies) (u, v) will there be spikes in the Fourier transform? Give a concrete formula for a notch filter $H(u, v)$ that will remove them.



$$g(x, y) = f(x, y) + \sin\left(\frac{2\pi}{10}x\right)$$

$$\dots \left(\because \sin\left(\frac{2\pi}{10}(x+10)\right) = \sin\left(\frac{2\pi}{10}x\right) \right)$$

$$G(u, v) = F(u, v) + \mathcal{F}\left(\sin\left(\frac{2\pi}{10}x\right)\right)$$

$$\therefore G(u, v) = F(u, v) + \frac{1}{2i} \left[\delta\left(u - \frac{1}{10}\right) - \delta\left(u + \frac{1}{10}\right) \right]$$

Yes, there will be spikes at $\left(\frac{1}{10}, 0\right), \left(-\frac{1}{10}, 0\right)$

We need a notch reject filter

example

$$H_1(u, v) = \begin{cases} 0 & D_1(u, v) \leq D_0 \\ 1 & \text{else} \end{cases}$$

$$D_1 = \sqrt{\left(u - \frac{1}{10}\right)^2 + v^2}$$

$$H_2(u, v) = \begin{cases} 0 & D_2(u, v) \leq D_0 \\ 1 & \text{else} \end{cases}$$

$$D_2 = \sqrt{\left(u + \frac{1}{10}\right)^2 + v^2}$$

$$H_{NR} = H_1 \cdot H_2$$

- (b) Now say the image has both periodic noise which you intend to remove using the notch filter above, and additive Gaussian noise which you intend to remove using a 5×5 mean filter. Will you get better results if you apply the notch filter first, or the mean filter first, or will you get identical results either way? Justify your answer.

$$g(x, y) = f(x, y) + \sin\left(\frac{2\pi}{10}x\right) + n(x, y)$$

$$\hat{f}(x, y) = g(x, y) * h_{na}(x, y) * w(x, y)$$

ideally we should get the exact same result on convolving mean filter ~~with~~ after notch filter & vice versa

Since

$$\hat{f}(x, y) = g(x, y) * h(x, y)$$

$$\hat{F}(u, v) = (G(u, v) \cdot H(u, v)) \cdot w(u, v) = (G(u, v) \cdot w(u, v)) \cdot H(u, v)$$

Some minute differences may occur due to ~~more~~ numeric approximations the peaks ~~is~~ due to sinusoid noise may cause numeric instabilities, hence it is better / safer to do notch filter then mean filter. ~~except for neighbourhood of~~ $(-\frac{1}{10}, 0)$ and $(\frac{1}{10}, 0)$ (in freq. domain) all other ~~non~~ values will be same.

5. Suppose you have a camera that always takes blurry and noisy photographs. The blur can be modeled as a fixed (but unknown) linear position-invariant degradation, while the noise is additive with a fixed (unknown) distribution. Design a procedure by which you can improve the quality of the images as much as possible. It may involve taking photographs of known scenes to calibrate the procedure, but after calibration, the procedure should be completely automated. Give full technical details of both the initial calibration steps, and the process for restoring a new photograph taken by the camera.

Assumptions:

- ① I have control over the control/known scenes
- ② $\eta \sim D(N)$: noise is from distribution D with
~~mean $\bar{\eta}$ and $E[\eta^2] = \bar{\eta}$~~
- ③ $g(x,y) = h(x,y) * f(x,y) + \eta(x,y)$

method:

- ① Take enough photos of pitch black scene/with lens cap to get multiple images, take avg over all pixels of all photos to estimate $E[\eta]$ and $E[\eta^2]$

since $f = 0$

$$g(x,y) = \eta(x,y)$$

and $E[\eta^2]$.. (info expected energy in fourier domain)

- ② Now take enough photos of a Dirac Delta scene i.e. pitch black with 1 pixel / small hole in center with maximum high intensity. take avg over all of them

[More space on next page...]

$$g(x, y) = h(x, y) * f(x, y) + n(x, y)$$

$$\therefore \bar{g}(x, y) = (h(x, y) * \bar{f}(x, y)) + \bar{n}(x, y)$$

$$\bar{f}(x, y) = \bar{g}(x, y)$$

$$\therefore \bar{g}(x, y) = h(x, y) + \bar{n}(x, y)$$

(3) from each pixel of any image remove $\bar{n}$ to get $h(x, y)$

Here on assume I have valid and correct h

$$(4) E[F^2] = \frac{E[G^2] - E[N^2]}{E[H^2]}$$

$$\therefore G = HF + N$$

$$G^* = \bar{H} \bar{F} + \bar{N}$$

$\Rightarrow$

$$G \cdot \bar{G} = H \bar{H} F \bar{F} + N \bar{N} + H F \bar{N} + \bar{H} \bar{F} N$$

assume F and N are uncorrelated.

$$\therefore E[G^2] = E[H^2] \cdot E[F^2] + E[N^2]$$

$$E[F \bar{N}], E[\bar{N} F] = 0$$

(5) I now have S_n (from calib) S_F (computed). ~~and~~

$H(x, y)$ (from calibration) apply Wiener filter

$$W(x, y) = \frac{\bar{H}(x, y)}{H^2(x, y) + (S_n/S_F)}$$

to ~~get~~ restore image.

Regularized deconvolution filters may also be used.

(Space for rough work)

$$x_0^{1-p} (1+p)(1+p^2) \dots (1+p^n)$$

$$f(x) = 0 < x_0$$

$$\frac{x_0}{1 - \frac{x_1 - x_0}{255}}$$

$$\frac{255}{x_1 - x_0} (x - x_0) < x_0$$

□

$$f = x^* y + \eta$$

by η

$$x - x_0 < \left(\frac{(x_1 - x_0) x_0}{255} \right) x_0$$

$$\sin \left(\frac{2\pi x}{M} \right)$$

$$= \frac{1 - e^{2\pi i \frac{x}{M}} - e^{-2\pi i \frac{x}{M}}}{2i} \quad 255 \frac{(x - x_0)}{x_1 - x_0} > x_1$$

$$\frac{S(u - \frac{1}{M}) - S(u + \frac{1}{M})}{2i}$$

$$x \geq \frac{x_1 (x_1 - x_0)}{255} + x_0$$

$$(0 \ 0 \ 0)$$

$$\{ \{ w(s, t) \delta(x - s, y - t) \}$$

$$w_1(0, 0) = 1$$

$$= J(x, y)$$

$$\begin{pmatrix} x & 0 & 0 \\ 0 & x & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$25x + 24y$$